Operations Research

Nonlinear optimization

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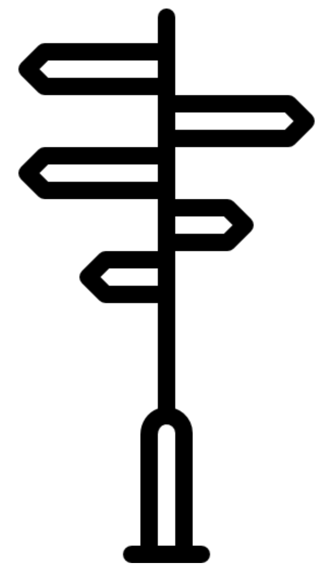
Course outline



- **Linear Optimization**
  - Modeling by linear programming and graphical solution
  - Solving linear programs, convexity
  - Simplex algorithm, 2-phase simplex
  - Simplex algorithm in matrix notation
  - Sensitivity analysis, reoptimization
  - Duality theory, zero-sum games

- **Integer Linear Optimization**
  - Modeling integer programming problems (IPs)
  - Solving IPs: branch & bound, cutting planes
  - Advanced solution methods: column generation, approximation algorithms
  - Fast solvable integer problems: total unimodularity, matroids
  - Network flow problems
  - Traveling salesperson problem

- **Nonlinear Optimization**
  - **Optimization of multidimensional functions**
  - Convex optimization

Today you will learn...

- basic terms and definitions related to **multivariate calculus** and **unconstrained nonlinear optimization**.

- how to characterize **extreme points** in multidimensional optimization problems.

- how to solve convex optimization problems using **gradient descent**.

- The initial slides provide a recap of your calculus class.

Definitions (recap)

- X: set of feasible solutions
- A set $X \subseteq \mathbb{R}^n$ is called **open** if for any $x \in X$ there exists $\varepsilon > 0$, such that $\mathcal{B}_\epsilon(x) = \{y \in \mathbb{R}^n | \|y - x\| < \varepsilon\} \subseteq X$. Example: $(0,1)$
- A **bounded** set is contained in $\mathcal{B}_M(0)$ for some M: $[2, \infty)$ is not bounded.
- A **closed** set contains the points at the boundary of the set.
- A **compact** set is closed and bounded.
- A set $X \subseteq \mathbb{R}^n$ is called **convex** if for all $x, y \in X$ any point $z = \lambda x + (1 - \lambda)y$ with $0 \leq \lambda \leq 1$ is in X.

Norms (recap)

Definition (Euclidean Norm):

In the n-dimensional Euclidean space $\mathbb{R}^n$, the Euclidean norm of a vector $x = (x_1, x_2, \cdots, x_n) \in \mathbb{R}^n$ is the length of the vector, computed as $\|x\|_2 = \sqrt{(x_1^2 + x_2^2 + \cdots + x_n^2)}$ or $\|x\|_2 = \sqrt{x^t x} = \sqrt{\langle x, x \rangle}$.

Sometimes the subscript is omitted for simplicity, and we write $\|x\|$.

Definition ($\boldsymbol{p}$-norm):

Let $p \geq 1$ be a real number. The p-norm of the vector $x \in \mathbb{R}^n$ is defined as

$\|x\|_p = \left(\sum_{i=1}^{n} |x|^p\right)^{1/p}$.

Nonlinear optimization

General form:

$$
\begin{array}{lll}
\text{max/min} & f(x) & \\
\text{s.t.} & g_i(x) \begin{Bmatrix} \geq \\ = \\ \leq \end{Bmatrix} 0 & \mathrm{i} = 1, \ldots, \mathrm{m} \\
 & x \in \mathbb{R}^n &
\end{array}
$$

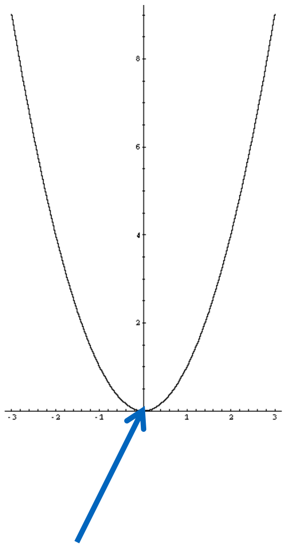
- In linear optimization, f and g_i are linear functions, but in the following they can also be nonlinear functions.
- We consider only real values (no integrality).
- An optimization problem involves maximizing/minimizing a function $f: X \to \mathbb{R}$.
- If $X = \mathbb{R}^n$ (no additional constraints), the problem is **unconstrained**.

Minimum vs. infimum

Example: $X \subset \mathbb{R}^n$ is a nonempty, convex set and $f(x) = x^2$.

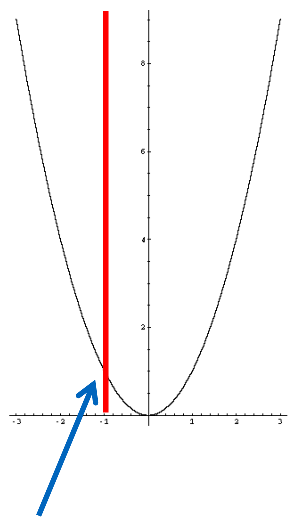
$$f'(x) = 2x,\ f''(x) = 2$$

$X = \mathbb{R}$



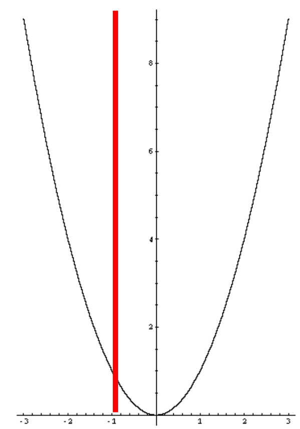
Minimum at $x = 0$
$(f'(x) = 0, f''(x) > 0)$

$X = \{x \in \mathbb{R}, x \leq -1\}$



Minimum at the boundary
at $x = -1$ $(f'(x) \neq 0)$

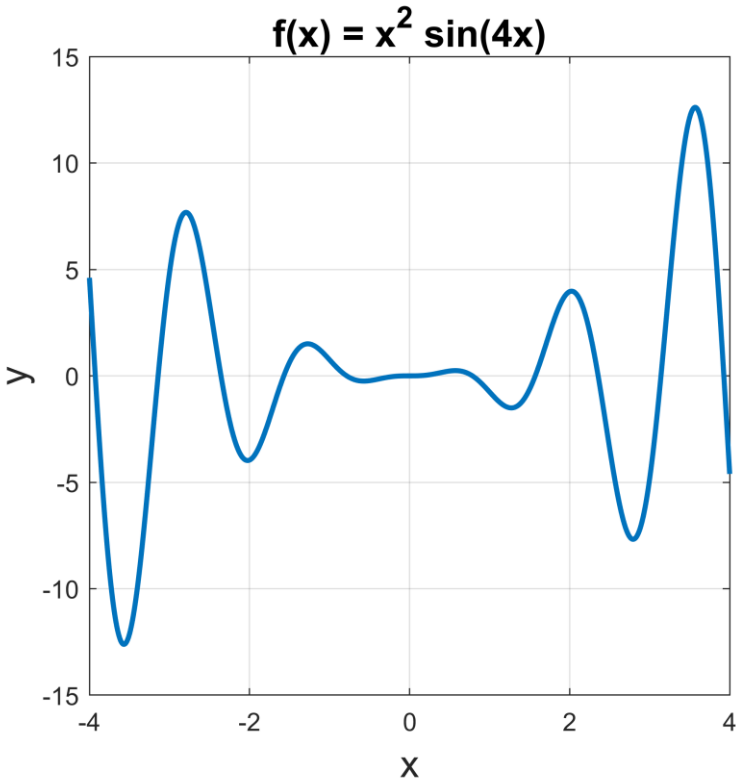
$X = \{x \in \mathbb{R}, x < -1\}$

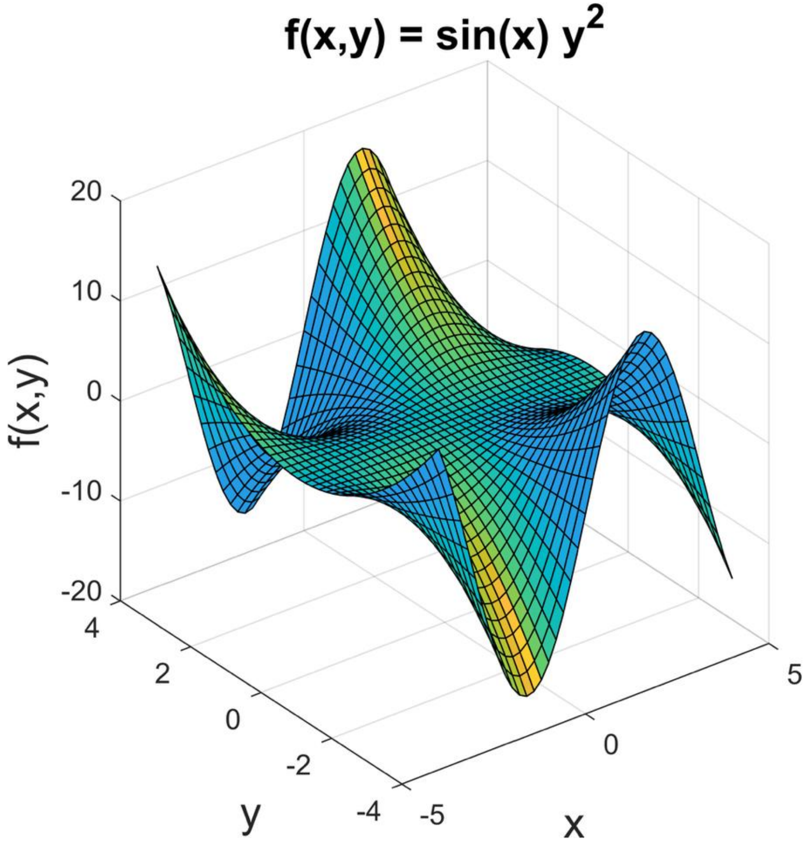


No minimum,
but an **infimum**

The infimum denotes the largest lower bound of a set. The minimum must be part of the set!

Nonlinear functions

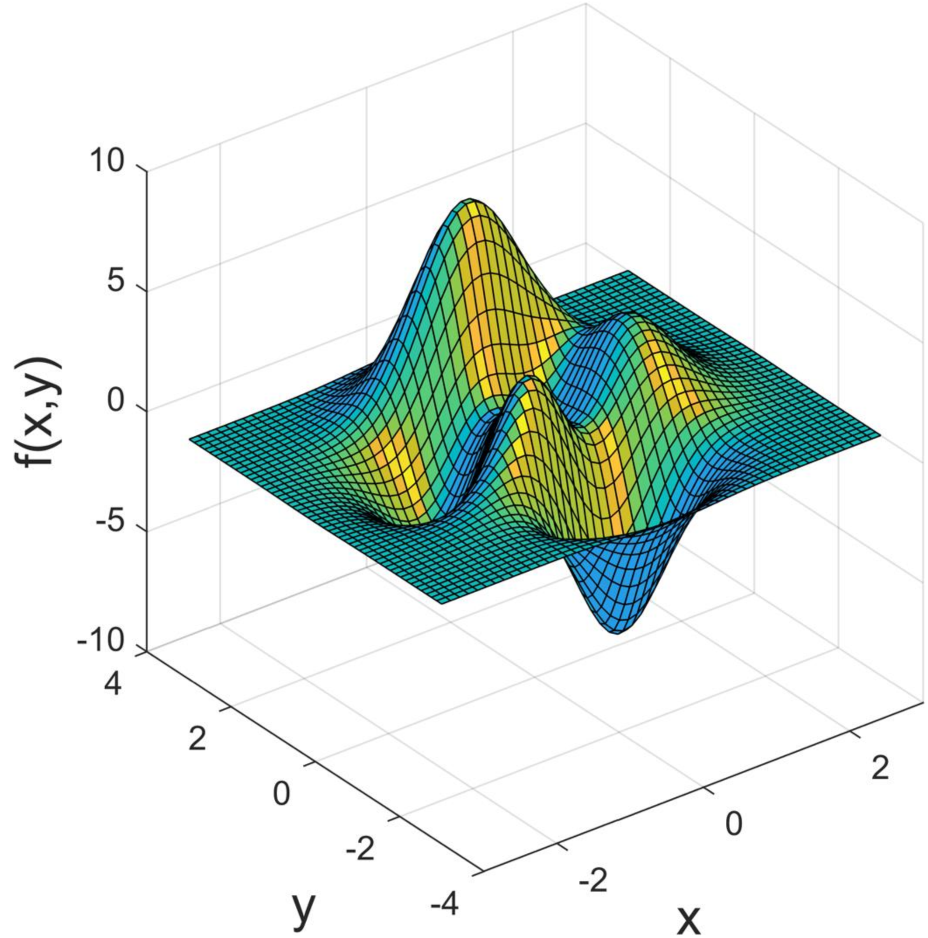




Extreme points

If $f(x') \geq f(x)$ for all $x \in X$, then $f(x')$ is a **global maximum** and x' is a **global maximum point**.

If there exists $\varepsilon > 0$ such that $f(x') \geq f(x)$ for all $x \in X$ with $\|x - x'\| < \varepsilon$, then $f(x')$ is a **local maximum** and x' is a **local maximum point**.



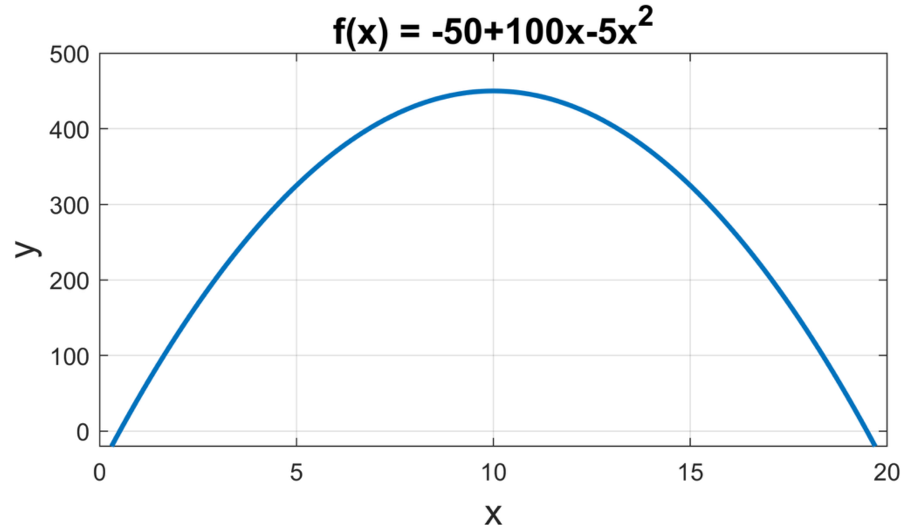
Optimality conditions for one variable (recap)

Necessary condition:

- $f'(x) = df/dx = 0$

Sufficient condition:

- $d^2f/dx^2 > 0$ minimum
- $d^2f/dx^2 < 0$ maximum



Necessary condition:

- $f(x) = y = -50 + 100x - 5x^2$
- $df/dx = 100 - 10x = 0, x = 10$

Example

Sufficient condition:

- $d^2f/dx^2 = -10$
  - since $d^2f/dx^2 < 0$: the maximum is at $x = 10$

Simple derivatives and univariate calculus

To find the places where the derivative is equal to zero, we can employ the standard derivative rules from calculus:

- $\frac{d}{dx}x^p = px^{p-1}$ for any $p \neq 0$
- $\frac{d}{dx}e^x = \frac{d}{dx}\exp(x) = e^x = \exp(x)$
- $\frac{d}{dx}\ln(\mathrm{x}) = \frac{1}{\mathrm{x}}$
- $\frac{d}{dx}f\big(g(x)\big) = f'\big(g(x)\big)g'(x)$
- $\frac{d}{dx}f(x)g(x) = f'(x)g(x) + f(x)g'(x)$

Partial derivatives

The **partial derivative** of a multivariate function f with respect to x_i is the derivative of f with respect to x_i assuming that all of the remaining variables are treated as constants

- $\frac{\partial}{\partial x}(x^2 y) = 2xy$
- $\frac{\partial}{\partial x}\left(y + \frac{x}{y}\right) = \frac{1}{y}$
- $\frac{\partial^2}{\partial x \partial y}(x^2 y) = \frac{\partial}{\partial y} 2xy = 2x = \frac{\partial}{\partial x} x^2$

Multidimensional optimization problems

Objective: Given $f: \mathbb{R}^n \to \mathbb{R}$.

Find $x^* = \underset{x \in \mathbb{R}^n}{\operatorname{argmin}} f(x)$

If $f: \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable, then the **gradient** at point x is defined as the vector of partial derivatives

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f(x)}{\partial x_1} \\ \frac{\partial f(x)}{\partial x_2} \\ \vdots \\ \frac{\partial f(x)}{\partial x_n} \end{pmatrix}$$

The vector $\nabla f(x)$ of partial derivatives $\frac{\partial f(x)}{\partial x_i}$ at x is the **gradient of f** and points in the direction of the steepest ascent.

Therefore, $-\nabla f(x)$ is a **descent direction**, at least for small $\eta > 0$ it holds:

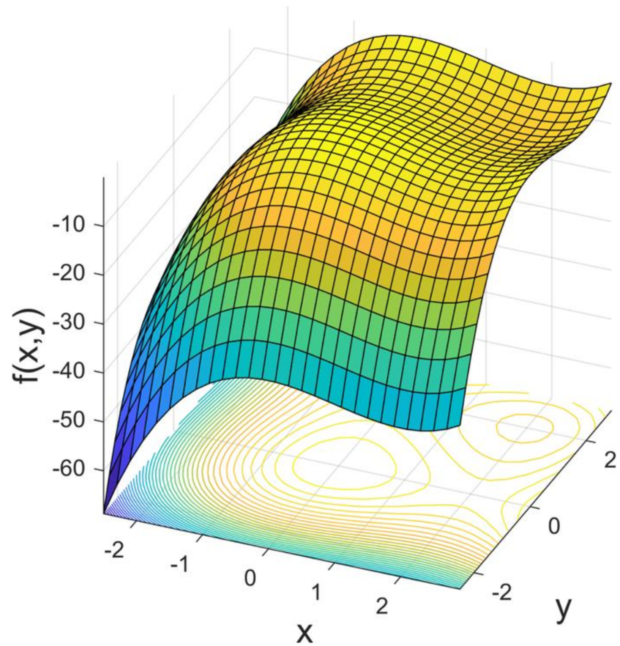
$$f\big(x - \eta \nabla f(x)\big) \leq f(x)$$

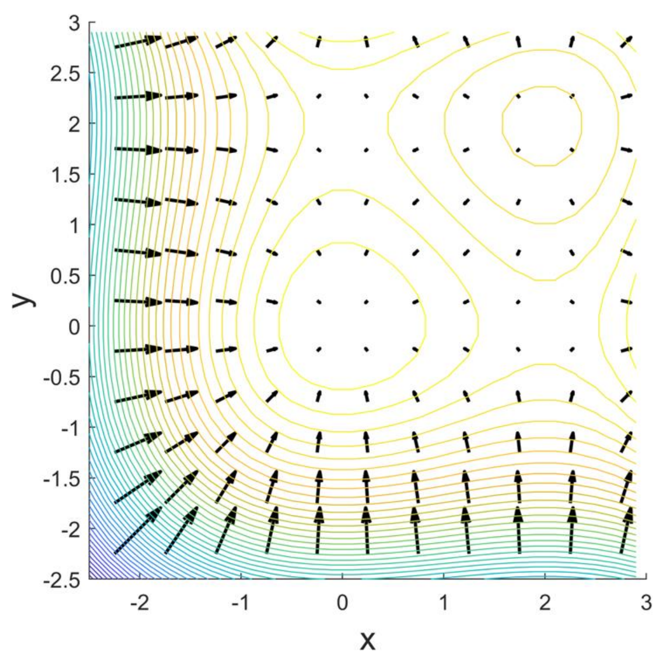
Example - gradient

$f(x, y) = x^3 - 3x^2 + y^3 - 3y^2$

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} = \begin{pmatrix} 3x^2 - 6x \\ 3y^2 - 6y \end{pmatrix}$$

- stationary (critical) points are at $3x^2 - 6x = 0$ and $3y^2 - 6y = 0$
- four critical points: $(0,0), (2,0), (0,2), (2,2)$
- a **directional derivative** indicates the rate of increase in a particular direction v: $\nabla f(x) v$





Directional Derivatives

The **directional derivative** of a differentiable function $f\colon \mathbb{R}^n \to \mathbb{R}$ at $\hat{x} \in \mathbb{R}^n$ along the vector $v \in \mathbb{R}^n$ is given by $\nabla f(\hat{x})^T v$

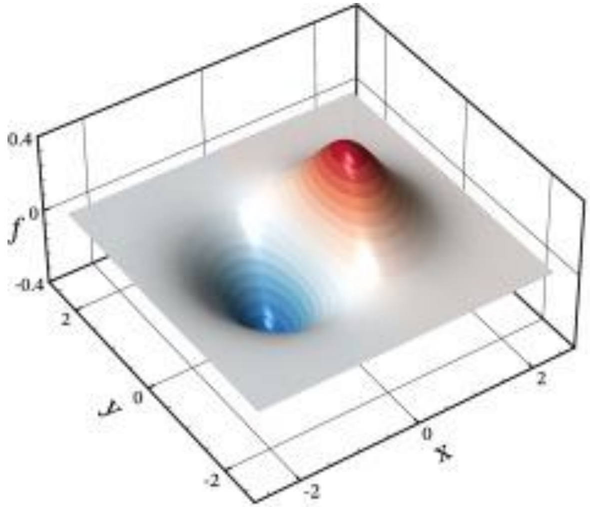
– Example: $f(x) = x_1^2 + 2x_1x_2 + x_2^2$, $v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\nabla f(\hat{x})^T v = 4\hat{x}_1 + 4\hat{x}_2$

If the directional derivative of a differentiable function $f\colon \mathbb{R}^n \to \mathbb{R}$ at $\hat{x} \in \mathbb{R}^n$ in the direction $v \in \mathbb{R}^n$ is equal to zero for all $v \in \mathbb{R}^n$, then we say that x is a **critical point**

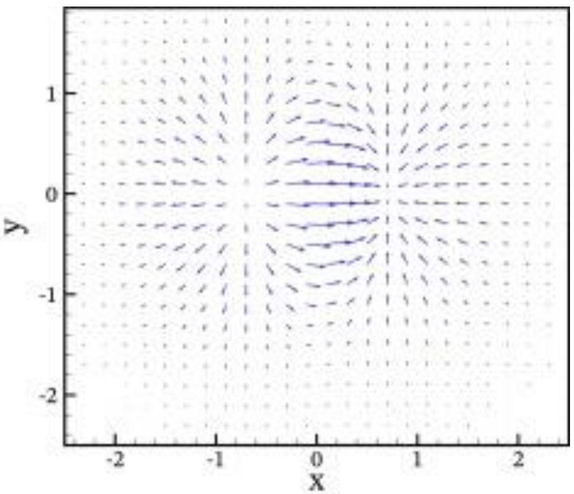
- This can only happen when $\nabla f(\hat{x}) = \vec{0}$
- Local maxima and minima of differentiable functions can only occur at places in which the gradient is zero for differentiable functions at interior points of an unconstrained domain.

Scalar fields vs. vector fields

- A **scalar field** is a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ that assigns a value to every point in a continuous space $\mathbb{R}^n$ (e.g., temperature or cost).
- A **vector field** is a function that assigns a vector to every point (e.g., Wind, Gradient).
- A **potential function** is a specific type of scalar field, whose gradient generates a specific vector field, the gradient field.
- The **gradient field** is the derivative ∇f of a potential function.



Scalar field



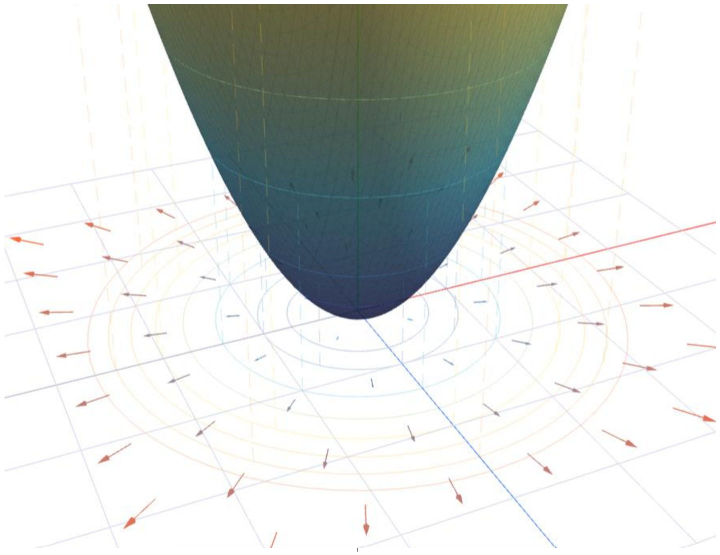
Gradient field

Divergence

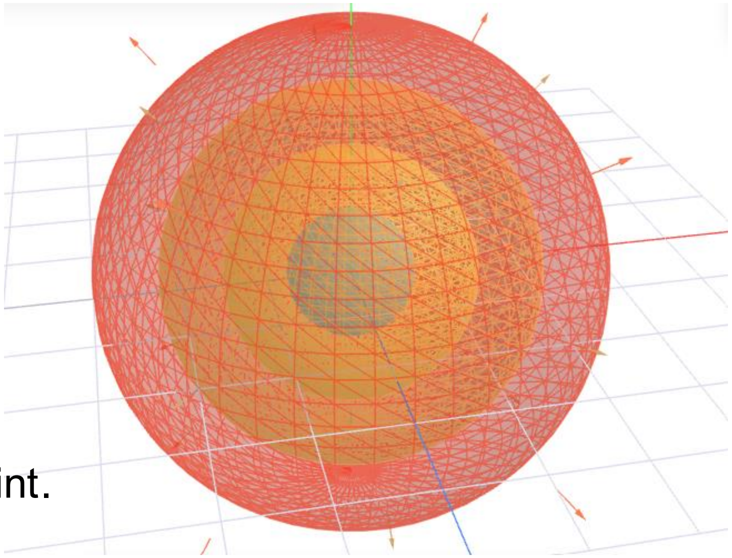
- Divergence takes a vector field (arrows) as input and returns a scalar (a single number) as output as a measure of expansion.
- This represents instability. The more you have, the faster you want to move away from having zero.

- You have a vector field $F(x, y) = \langle P, Q \rangle$, where P is the flow in the x-direction and Q is the flow in the y direction.
  - $\mathrm{div}\, F = \nabla \cdot F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}$

- Divergence of a gradient tells you if the arrows are converging to a peak or a valley
  - If $\mathrm{div}(\nabla f) > 0$, the arrows point outward (min)
  - If $\mathrm{div}(\nabla f) < 0$, the arrows point inward (max)



Operations of vector fields (3D)



Let's look at the scalar potential $\phi(x,y,z) = x^2 + y^2 + z^2$

The gradient field is $\nabla\phi(x,y,z) = (2x, 2y, 2z)$
The induced vector field is $F(x,y,z) = (2x, 2y, 2z)$
$F_1 = 2x, F_2 = 2y, F_3 = 2z$

Divergence:
measures how much a vector field spreads out from a point.

$$\nabla \cdot F = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = 2 + 2 + 2 = 6$$

Curl:
measures the local rotation of a vector field

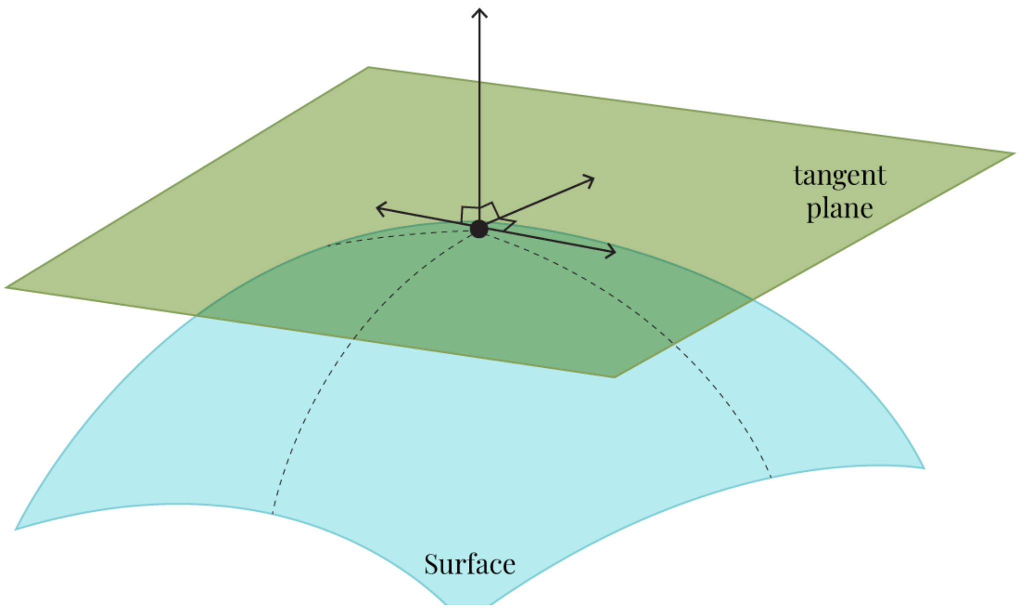
$$\nabla \times F = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) = (0,0,0)$$

Curl is just discussed for completeness, and is not important for this class.

Tangent plane

- Complex non-linear functions are hard to solve. Linear functions (planes) are easy to solve.
- The tangent plane is the flat plane that best approximates a curved surface at a point.
- $L(x, y)$ is the linearization function. The tangent plane is the geometric graph of that function.

$$L(x, y) \approx f(x_0, y_0) + \nabla f(x_0, y_0)^T \cdot \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix}$$



Extreme points given multiple variables

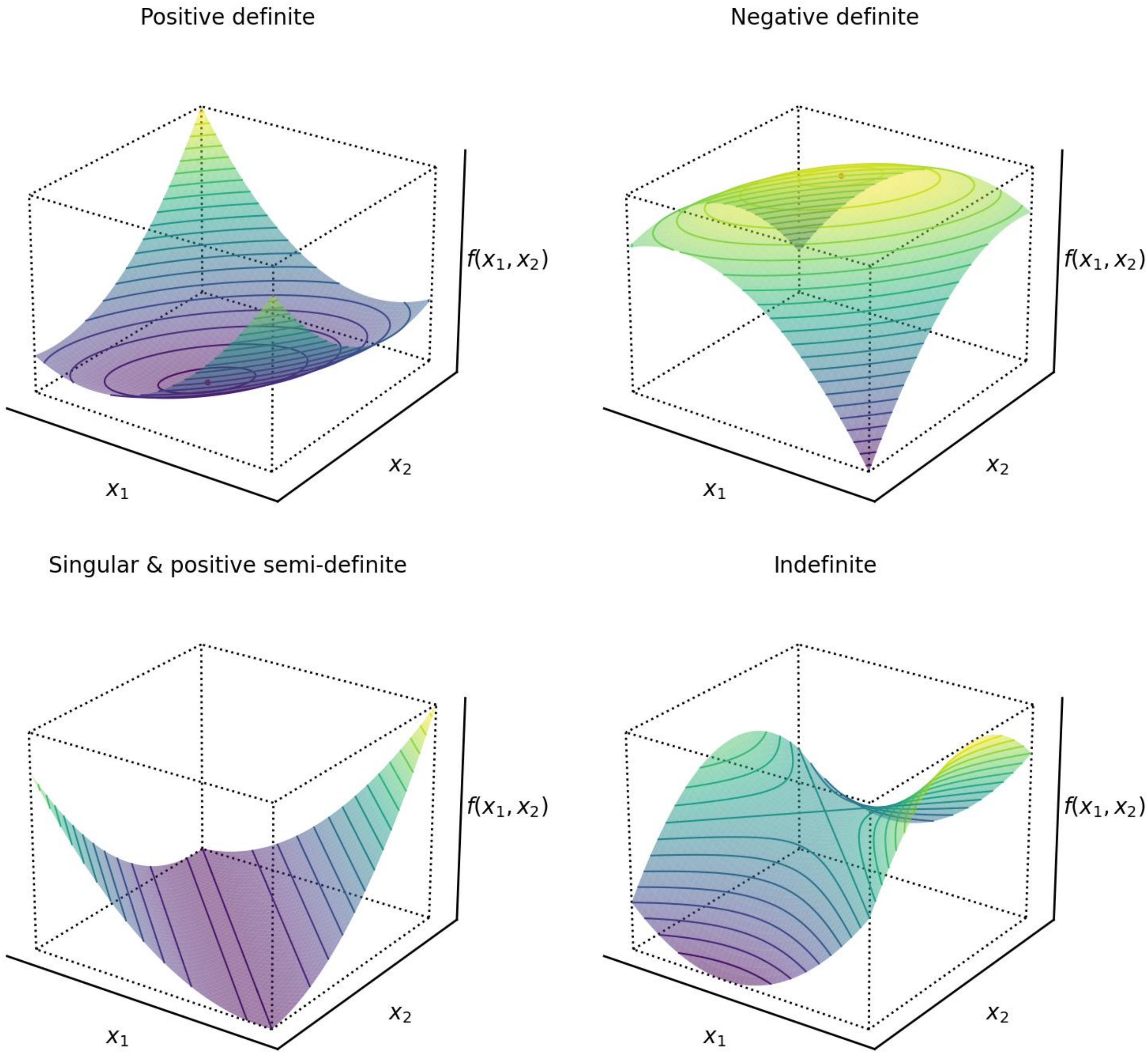
Let $X \subseteq \mathbb{R}^n$ be open and $f: X \to \mathbb{R}$ a twice continuously differentiable function for $x = (x_1, \dots, x_n)$. Then:

- $\nabla f(x') = 0$ is a necessary condition for x' to be a local maximum or minimum point, respectively.
- $\nabla f(x') = 0$ and $H(x')$ positive definite ($H(x')$ negative definite, resp.) is a sufficient condition for x' to be a local minimum (maximum) point.

- $H(x')$ negative definite ($-H(x')$ positive definite, resp.) → maximum ($-x^2$)
- $H(x')$ positive definite, i.e. $x^T H x > 0$ → minimum (x^2)
- $H(x')$ indefinite → saddle point (e.g., $f(x, y) = x^2 - y^2$)
- $H(x')$ positive/negative semidefinite → max, min or saddle (-x^4, x^4, x^3, $x' = 0$)
- If the Hessian matrix is positive semidefinite everywhere on X → convexity

Tests for definiteness

Type of definiteness	Condition
Positive definite	All eigenvalues > 0, all leading principal minors > 0
Negative definite	All eigenvalues < 0, principal minors alternate in sign (starting with < 0)
Indefinite	Eigenvalues of mixed sign
Positive semidefinite	All eigenvalues >= 0
Negative semidefinite	All eigenvalues <= 0



Source: https://gregorygundersen.com/blog/2022/02/27/positive-definite/

Example: positive definiteness via eigenvalues

$$\min x_1{}^2 + 3x_2{}^2 + x_1x_2 - 3x_1 - 7x_2$$

$$\nabla f(x) = (2\,x_1 + x_2 - 3, x_1 + 6x_2 - 7) = 0 \Rightarrow x_1 = 1, x_2 = 1$$

$$H = \begin{pmatrix} 2 & 1 \\ 1 & 6 \end{pmatrix}$$

$Hv = \lambda v$ positive λ => scaling

$(H - \lambda I)v = 0$

$|H - \lambda I| = 0$

$$|H - \lambda I| = \begin{vmatrix} 2-\lambda & 1 \\ 1 & 6-\lambda \end{vmatrix}$$

$$= (2-\lambda)(6-\lambda) - 1 = 0$$

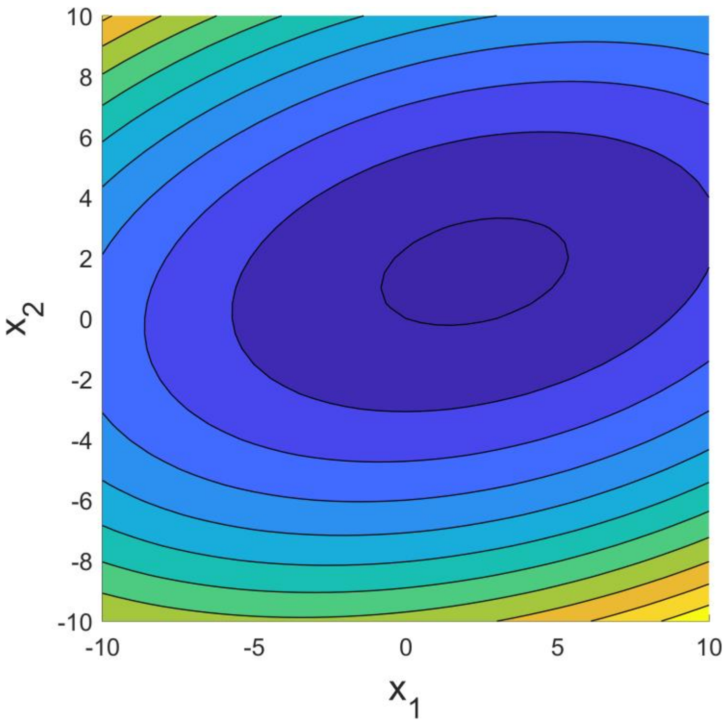
$\Rightarrow \lambda^2 - 8\lambda + 11 = 0$

$\Rightarrow \lambda = 4 \pm \sqrt{5} > 0$

$\Rightarrow$ positive definite

Remember

$$\frac{-b + \sqrt{b^2 - 4ac}}{2a}$$



$\Rightarrow$ **convex** and **minimum**

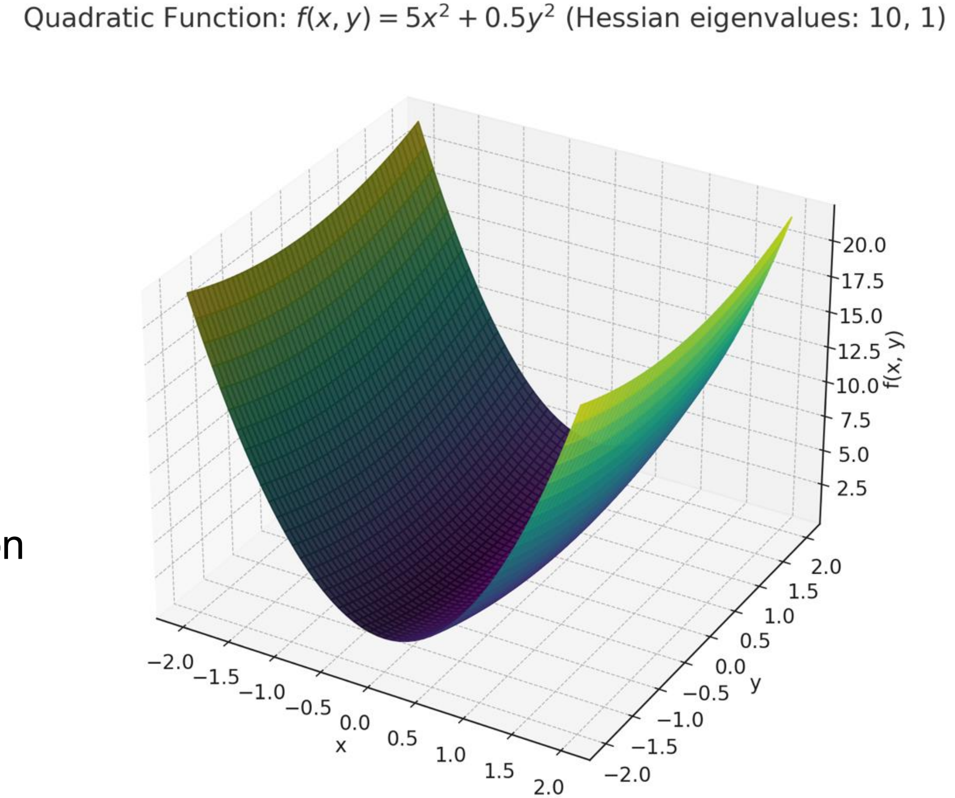
Eigenvalues of a Hessian

$f(x, y) = 1/2\ (x, y)^T Q(x, y)$

$\mathrm{Q} = \begin{pmatrix} 10 & 0 \\ 0 & 1 \end{pmatrix}$

$\lambda_1 = 10, \lambda_2 = 1$

A large $|\lambda_i|$ => very curved in this direction



Nonlinear unconstrained optimization

<u>Definiteness of a matrix:</u>

A symmetric matrix H is called **positive *(semi-)*definite** if $x^T H x > 0 \ (\geq 0)$ for all $x \neq 0$.

Theorem: A symmetric (Hessian) matrix H is positive definite if and only if all leading principal minors are positive:

$$H_1 = |h_{11}|$$

$$H_2 = \left| \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \right| = h_{11} \cdot h_{22} - h_{12} \cdot h_{21}$$

$$\ldots$$

$$H_n = |H|$$

Theorem: A symmetric matrix H is positive definite if and only if all eigenvalues are positive.

Principal minors

Let A be a symmetric $n \times n$ matrix. We can determine the definiteness of A by calculating the eigenvalues or the principal minors.

Definition:
If minors are created by deleting rows and columns of the same numbers, they are called **principal minors**, or more **precisely principal minors of kth order**, if the size of the submatrix is to be specified. If exactly the first k rows and columns remain, we call these the **leading principal minors of kth order**.

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

- Leading principal minor of order 1: 1
- Leading principal minor of order 2: -3

Note: Despite all entries of A being non-negative, A is not positive definite.

Example

$$f(x,y) = x^3 - 3x^2 + y^3 - 3y^2$$

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} = \begin{pmatrix} 3x^2 - 6x \\ 3y^2 - 6y \end{pmatrix}$$

$$H(x,y) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}$$

$$= \begin{pmatrix} 6x - 6 & 0 \\ 0 & 6y - 6 \end{pmatrix}$$

Rules

(a) A is positive definite $\Leftrightarrow D_k > 0$ for all leading principal minors.

(b) A is negative definite $\Leftrightarrow (-1)^k D_k > 0$ for all leading principal minors.

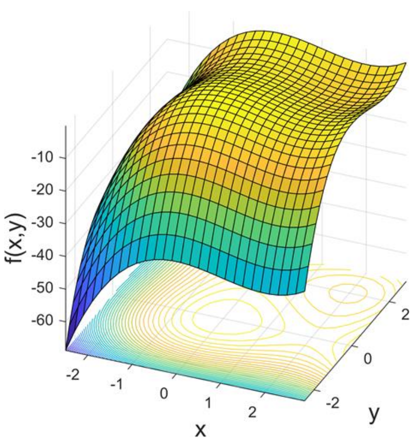
(c) If none of the leading principal minors is 0 and neither (a) nor (b) hold, the matrix is indefinite.

at (0,0): $\begin{pmatrix} -6 & 0 \\ 0 & -6 \end{pmatrix}$ is negative definite ($D_1 = -6, D_2 = 36$), maximum.

at (2,0): $\begin{pmatrix} 6 & 0 \\ 0 & -6 \end{pmatrix}$ is indefinite ($D_1 = 6, D_2 = -36$), saddle point.

at (0,2): $\begin{pmatrix} -6 & 0 \\ 0 & 6 \end{pmatrix}$ is indefinite ($D_1 = -6, D_2 = -36$), saddle point.

at (2,2): $\begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix}$ is positive definite ($D_1 = 6, D_2 = 36$), minimum.



Example: indefiniteness

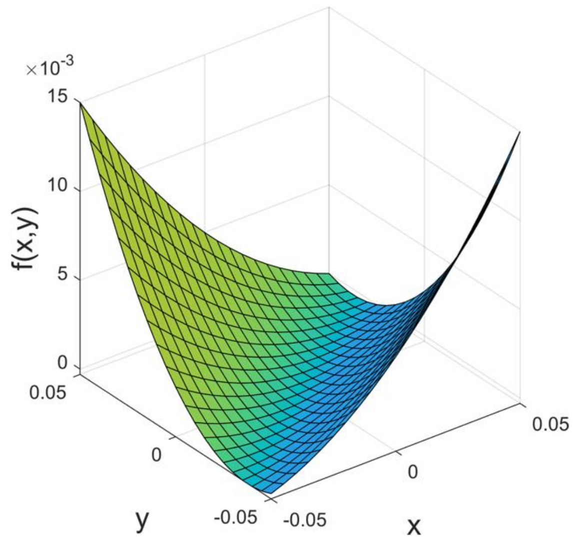
$f(x, y) = x^2 - 3xy + 2y^2$

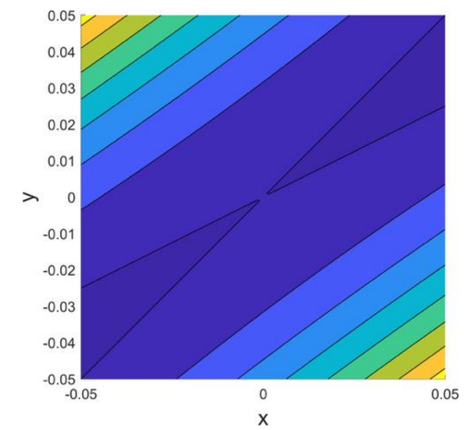
$\frac{\partial f}{\partial x} = 2x - 3y, \frac{\partial f}{\partial y} = -3x + 4y, \frac{\partial^2 f}{\partial x \partial y} = -3, \frac{\partial^2 f}{\partial x \partial x} = 2, \frac{\partial^2 f}{\partial y \partial y} = 4$

Gradient: $\nabla f(x, y) = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$

Hessian matrix:

$$H(x) = \begin{pmatrix} 2 & -3 \\ -3 & 4 \end{pmatrix}$$





The first leading principal minor is 2 (positive).

The second leading principal minor is $2 \cdot 4 - (-3) \cdot (-3) = -1 < 0$, so not positive.

Neither a maximum nor a minimum, but a saddle point.

Summary

The vector $\nabla f(x)$ of partial derivatives $\frac{\partial f(x)}{\partial x_i}$ at the point x is called **gradient** of f (it points at each point in the **direction of steepest ascent**).

The symmetric matrix $H(x) = \left(\frac{\partial f^2(x)}{\partial x_i \partial x_j}\right)_{\mathrm{i,j}}$ is called the **Hessian matrix.**

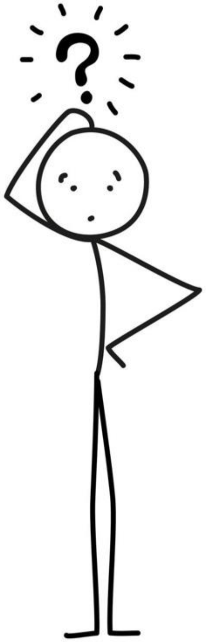
A symmetric matrix A is called **positive definite** if $x^t A x > 0$ for all $x \neq 0$
(all eigenvalues are >0).

A symmetric matrix A is called **negative definite** if $x^t A x < 0$ for all $x \neq 0$
(all eigenvalues are <0).

A symmetric matrix A is called **indefinite** if the eigenvalues are both >0 and <0.

A symmetric matrix A is called **positive semidefinite** if $x^t A x \geq 0$ for all $x \neq 0$
(all eigenvalues are ≥0).

Do you remember the definition of convexity in the context of linear optimization?

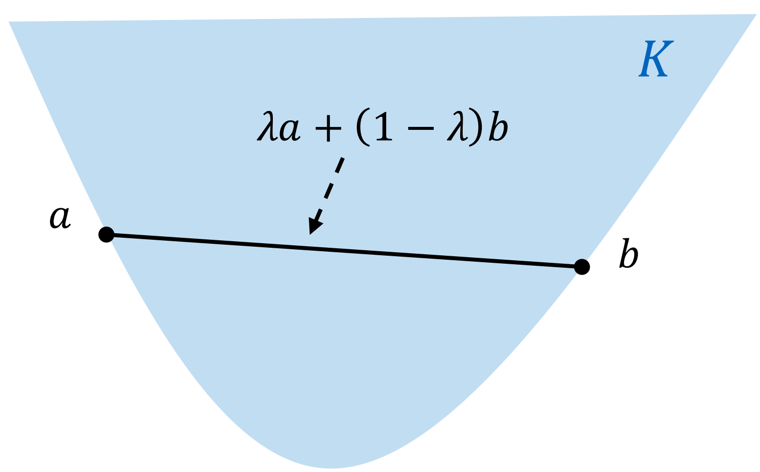


Convex sets (visualized)

A **set** K is called **convex** if for any two points $a, b \in K$ the point

$$x = \lambda a + (1 - \lambda) b$$

with $0 \leq \lambda \leq 1$ is in K.



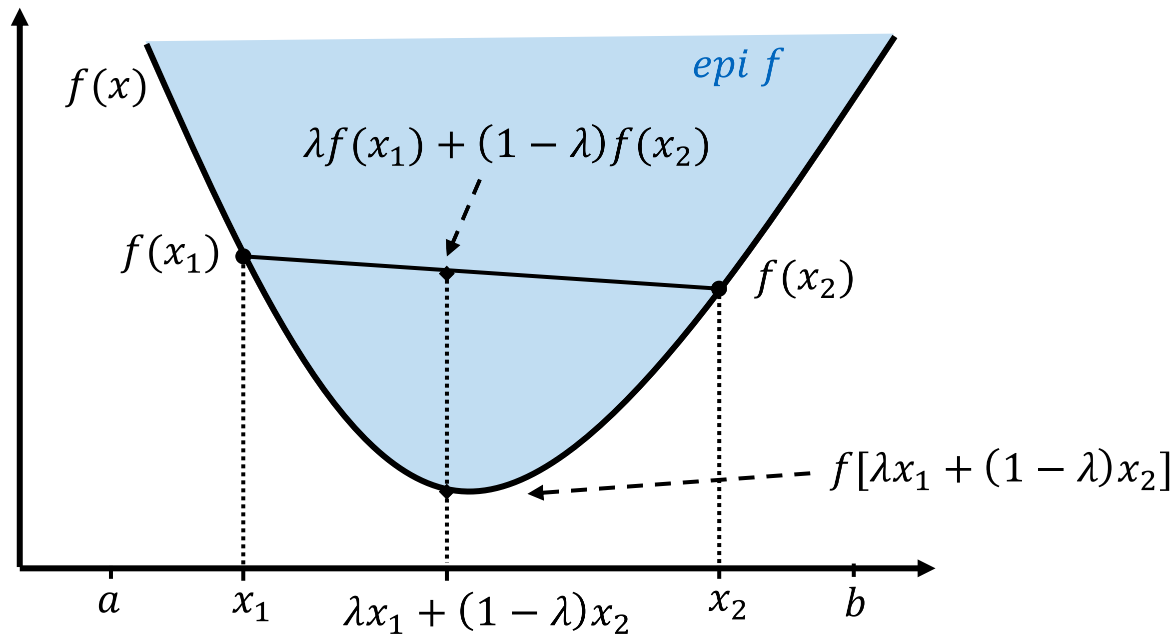
Convex functions

A **function** f is **convex** at an interval (a, b) if for any $x_1, x_2 \in (a, b)$ and $0 \leq \lambda \leq 1$:

$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda) f(x_2)$$

A function f is **strictly convex** for all $x_1 \neq x_2$ if the equality holds only for $\lambda = 0$ or $\lambda = 1$ and $0 \leq \lambda \leq 1$.

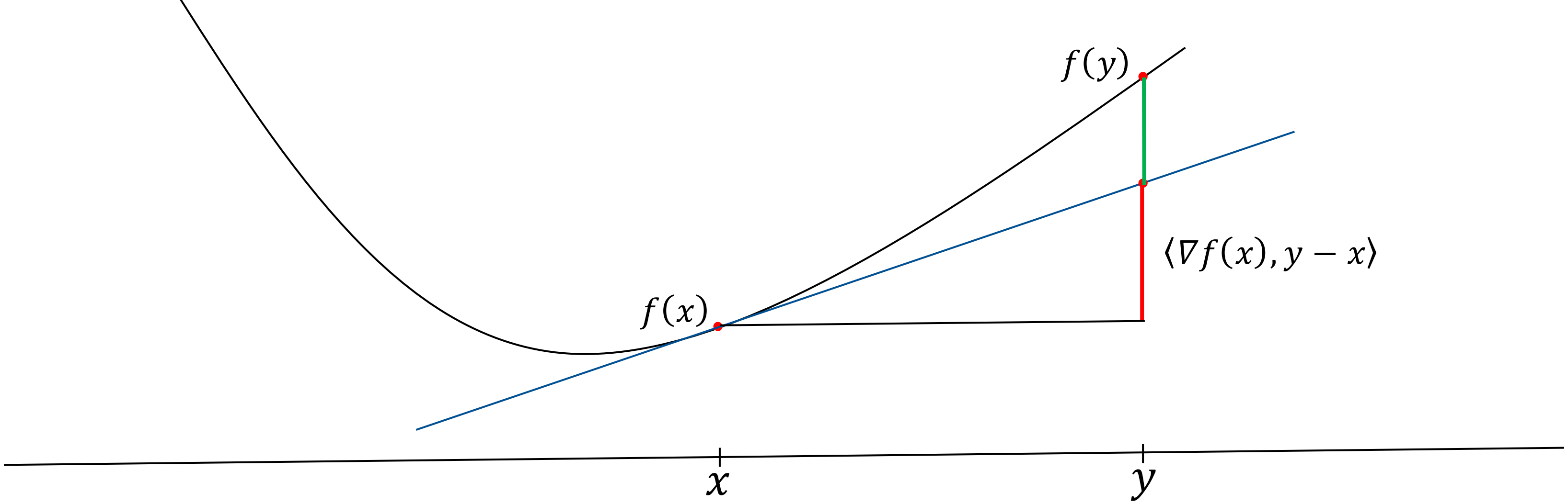
Epigraph of f:



Convexity of differentiable functions

- A differentiable function f is convex if

$$f(y) - f(x) \geq \langle \nabla f(x), y - x \rangle$$



- Examples:
  - The function $f(x) = x^2$ is **strictly convex.**
  - The function $f(x) = |x|$ is **convex**, but not strictly convex.

Strong convexity and smoothness

- A differentiable function f is a **strictly convex**, if

$$f(x) - f(y) > \langle \nabla f(y), x - y \rangle$$

  Strictly convex functions ensure that the function lies strictly below the line segment connecting any two points.

- A differentiable function f is **α-strongly convex** w.r.t. an arbitrary norm $\|\cdot\|$, if

$$f(x) - f(y) \geq \langle \nabla f(y), x - y \rangle + \frac{\alpha}{2} \|x - y\|^2$$

  Strongly convex functions ensure that the function lies below the line segment by a specific quadratic term, indicating a stronger degree of curvature and uniqueness of the minimum.

- A differentiable function f is **β-smooth** w.r.t. an arbitrary norm $\|\cdot\|$, if

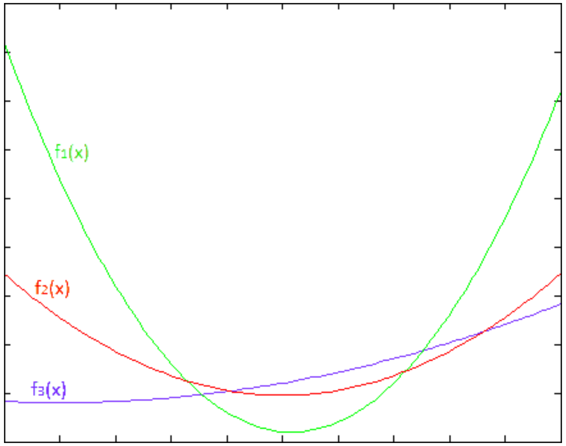
$$\|\nabla f(x) - \nabla f(y)\|_* \leq \beta \|x - y\|$$

  This means, the gradient does not change too fast at any point.
  $f(x) = |x|$ is convex but not smooth.

Transformations of convex functions

Examples of convex functions:

- linear functions: $f(x) = a^T x$
- affine functions: $f(x) = a^T x + b$
- exponential functions: $f(x) = e^{\alpha x}$
- norms: $f(x) = \|x\|$

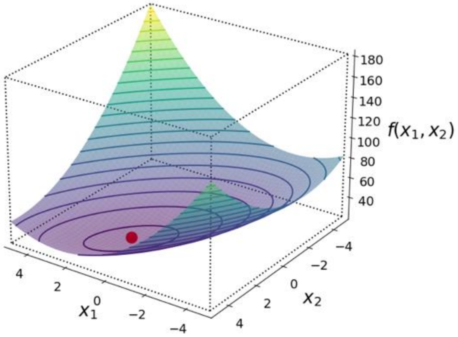


Transformations

- Non-negative multiples: αf is convex if f is convex and $\alpha \geq 0$.
- Sum: $f + g$ is convex if f, g are convex.
- Composition with affine functions: $f(Ax + b)$ is convex if f is convex (e.g., $f(x) = \|Ax + b\|$).
- $f(x) = h\big(g(x)\big)$ is convex if g is convex and h is convex and non-decreasing.
- $f(x) = \max\{f_1(x), \cdots, f_m(x)\}$ is convex if the components are convex.

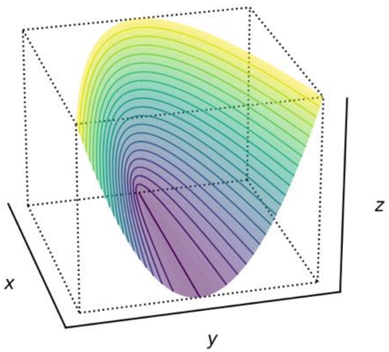
Positive semi/definite matrices

$\nabla^2 f(x)$ positive semidefinite $\forall x$ => convex function
$\nabla^2 f(x)$ positive definite $\forall x$ => strictly convex function
$\nabla^2 f(x)$ negative definite $\forall x$ => strictly concave function



If a symmetric matrix defines the **Hessian of a quadratic function** f, and if it is **positive semidefinite**, the function f is **convex**.

$f(x) = x^T Qx + bx + c$ (quadratic function with Q symmetric)
$\nabla f(x) = 2Qx + b$ (Gradient)
$\nabla^2 f(x) = 2\mathrm{Q}$ (Hessian)



A function with a **positive semidefinite Hessian** everywhere is convex.
For a quadratic function $f(x) = x^T Qx + bx + c$, this means that f is convex exactly when Q is positive semidefinite.

Convexity and monotonicity

Definition:
Let $X \subset \mathbb{R}^n$ be an open and convex set and $f: X \to \mathbb{R}$ continuously differentiable. Then:

a) f is **convex** (on X) if and only if for all $x, y \in X$ it holds:
$$f(x) - f(y) \geq \nabla f(y)^T (x - y) \ \forall x, y \in dom\, f$$
b) f **strictly convex** (on X) if and only if for all $x, y \in X$ it holds:
$$f(x) - f(y) > \nabla f(y)^T (x - y) \ \forall x \neq y$$

Definition:
Let $X \subset \mathbb{R}^n$. A function $g: X \to \mathbb{R}^n$,

a) is called **monotone** (on X) if for all $x, y \in X$ it holds:
$$(x - y)^T \big(g(x) - g(y)\big) \geq 0;$$
b) is called **strictly monotone** (on X) if for all $x, y \in X$ $(x \neq y)$ it holds:
$$(x - y)^T \big(g(x) - g(y)\big) > 0;$$

Implications

f strictly convex $\Leftrightarrow \nabla f$ strictly monotone $\Leftarrow \nabla^2 f$ positive definite

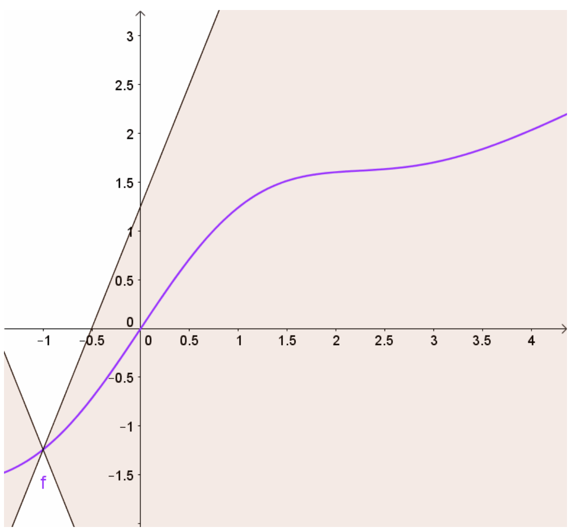
f convex $\Leftrightarrow \nabla f$ monotone $\Leftrightarrow \nabla^2 f$ positive semidefinite

Lipschitz continuous gradients

Let $\mathbb{V}$ be a [finite dimensional] vector space over $\mathbb{R}$, and $||\cdot||$ be an arbitrary norm on $\mathbb{V}$.
A function $f: \mathbb{V} \to \mathbb{R}$ is Lipschitz continuous with Lipschitz-constant L, if $\forall x, y \in \mathbb{V}$:

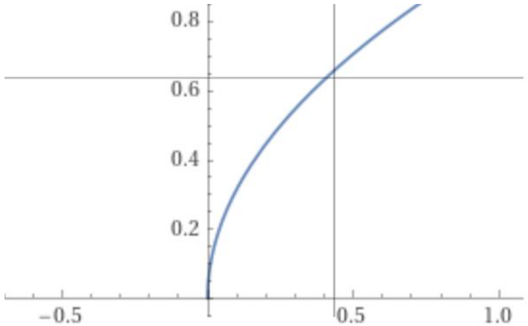
$$||f(x) - f(y)|| \leq L \cdot \|x - y\|$$

Geometrically, $f: \mathbb{V} \to \mathbb{R}$ is L-Lipschitz if there is a double-cone with steepness L, such that the graph of f always stays outside of the double-cone



Source: wikipedia.org

Violation of the Lipschitz condition:
$f(x) = \sqrt{x}$ on $[0,\infty)$
$f'(x) = 1/2\sqrt{x}$
$\left|\sqrt{x} - \sqrt{y}\right| / |x - y|$ is unbounded near 0



Summary: convexity

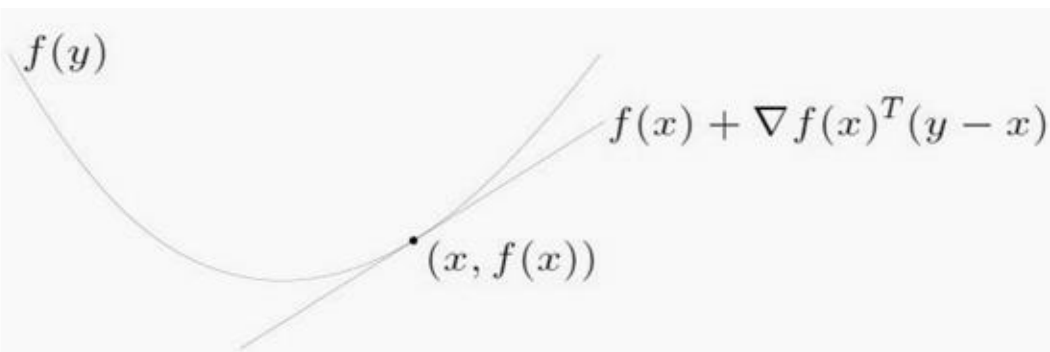
Characterizations using gradients

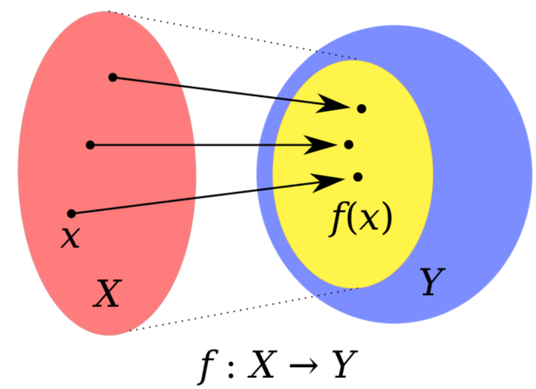
The gradient allows an alternative definition of convex functions:

A differentiable function f is convex if and only if dom f is also convex and

$f(y) - f(x) \geq \nabla f(x)^T(y - x)\ \forall x, y \in dom\ f$

$\equiv f(x) - f(y) \leq \nabla f(x)^T(x - y)\ \forall x, y \in dom\ f$





The tangent touches the function f at the point $(x, f(x))$.

The red ellipse is dom f.

Characterizations using the Hessian matrix:

A twice differentiable function f is convex (concave) if and only if dom f is convex and the Hessian matrix is positive semidefinite **everywhere**: $\nabla^2 f(x) \geq 0\ \forall x \in dom\ f$

Local and global maxima

Nonlinear optimization problems are, in general, difficult to solve.

Let $X \subseteq \mathbb{R}^n$ be convex and $f: X \rightarrow \mathbb{R}$ concave, then each local maximum is a global maximum.

Moreover, if X is open and f is twice continuously differentiable, then $\nabla f(x')$=0 is a sufficient condition for x' to be a global maximum point (for one variable, if $f'(x')$=0).

Unfortunately, it is not always easy to calculate the null points of the gradients analytically. Sometimes the functions are also not differentiable
→ numerical iterative methods to find extreme points.

For convex functions there are efficient numerical methods, even if the minimum is not easy to determine analytically!

Convex optimization

Definition (Convex optimization):

Let $X \subset \mathbb{R}^n$ be a convex set and $f: X \to \mathbb{R}$ a convex function.

Compute the minimum of the function f over the feasible set X: $\min_{x \in X} f(x)$.

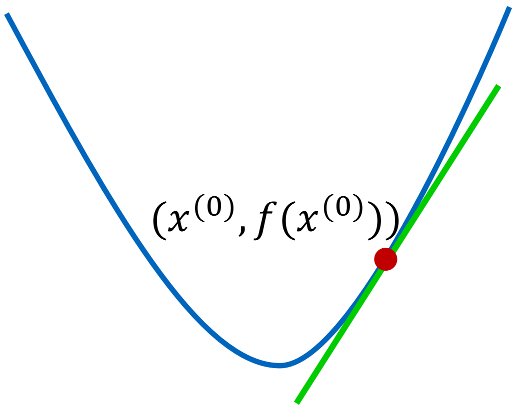
All local solutions of convex optimization problems are also global.

Special cases

- if $X = \mathbb{R}^n$ we talk about unconstrained optimization
- linear optimization is a special case of convex optimization
- $\exp x, -\log x, x \log x$
- norms and projections

Gradient descent method

Task: $\min f(x)\,, s.t.\, x \in\ X = \mathbb{R}^n$



Derivation via 1st order Taylor approximation
(i.e., via a linear function)
$f(x) \approx f\left(x^{(0)}\right) + \left\langle \nabla f\left(x^{(0)}\right), x - x^{(0)} \right\rangle$

This would follow the gradient arbitrarily far. However, the approximation is only good around $x^{(0)}$. Therefore, we add a penalty term and use the following formula, starting from $k = 0$.

$$x^{(k+1)} = \operatorname{argmin}_x f(x^{(k)}) + \left\langle \nabla f\left(x^{(k)}\right), x - x^{(k)} \right\rangle + \frac{1}{2\eta}\left\|x - x^{(k)}\right\|^2$$
$$= \operatorname{argmin}_x \left\langle \nabla f\left(x^{(k)}\right), x \right\rangle + \frac{1}{2\eta}\left\|x - x^{(k)}\right\|^2$$

Set first derivative of x to zero:

$$0 + \nabla f\left(x^{(k)}\right) + \frac{1}{\eta}\left(x - x^{(k)}\right) = 0$$
$$x^{(k+1)} = x^{(k)} - \eta \nabla f\left(x^{(k)}\right)$$

Gradient ascent method

- Greedy algorithm of the steepest ascent (gradient method) for solving problems with several variables
- Function must be continuously differentiable

Input: concave, continuously differentiable function $f\colon \mathbb{R}^n \to \mathbb{R}$,
feasible start point $x^{(1)} \in \mathbb{R}^n$ and parameter $\varepsilon > 0$

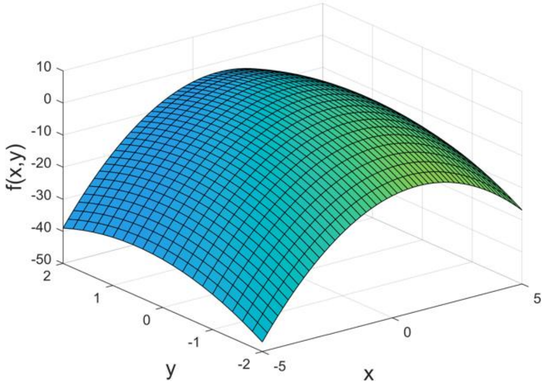
$k = 1$
While($\left\|\nabla f(x^{(k)})\right\| \geq \varepsilon$){
- Compute $\eta^* > 0$ such that $f\left(x^{(k)} + \eta \nabla f\left(x^{(k)}\right)\right)$ is maximal
- Set $x^{(k+1)} = x^{(k)} + \eta^* \nabla f\left(x^{(k)}\right)$
- $k++$

}

Gradient ascent method

$f(x_1, x_2)$: $2(x_1 + x_2) - x_1^2 - 2x_2^2$

$\nabla f(x) = \begin{pmatrix} 2(1 - x_1) \\ 2(1 - 2x_2) \end{pmatrix}$

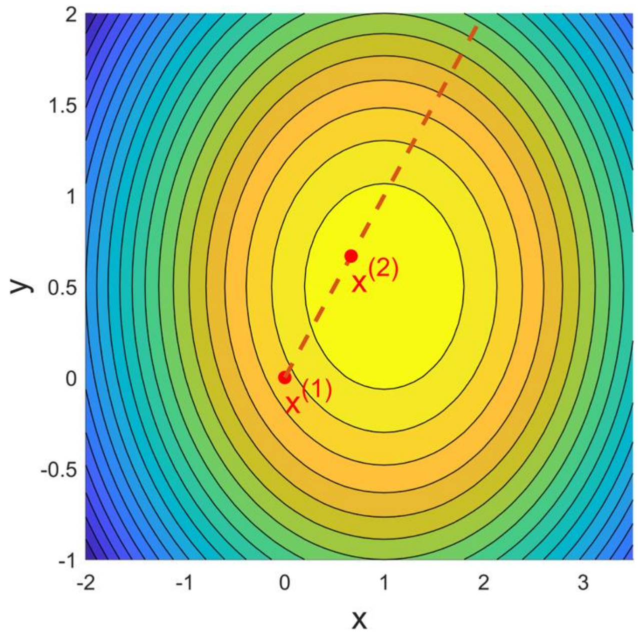


- Start at $x^{(1)} = (0{,}0), f\left(x^{(1)}\right) = 0, \nabla f\left(x^{(1)}\right) = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$
- Maximize $f(0 + 2\eta, 0 + 2\eta) = 8\eta - 12\eta^2$
  - Step η in the direction (line) of the gradient
- Set derivative to 0: $8 - 24\eta = 0$

→ maximum point is at $\frac{1}{3}$

→ $x^{(2)} = (\frac{2}{3}, \frac{2}{3}), f\left(x^{(2)}\right) = \frac{4}{3}, \nabla f\left(x^{(2)}\right) = \begin{pmatrix} \frac{2}{3} \\ -\frac{2}{3} \end{pmatrix}$

…



Momentum

Gradient descent:

1. Choose a start point $x^{(0)}$
2. Choose the maximum number of iterations T
3. Choose a step size $\eta \in [a, b]$
4. Repeat $x^{(k+1)} = x^{(k)} - \eta \nabla f\left(x^{(k)}\right)$

The gradient descent method forgets the history.
The function value can oscillate.
This leads to inefficiencies.

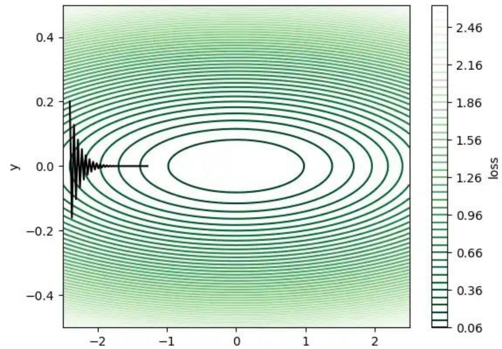
Momentum as weighted average of gradients :

$z^{(k+1)} = -\eta \nabla f\left(x^{(k)}\right) + \beta z^{(k)}$

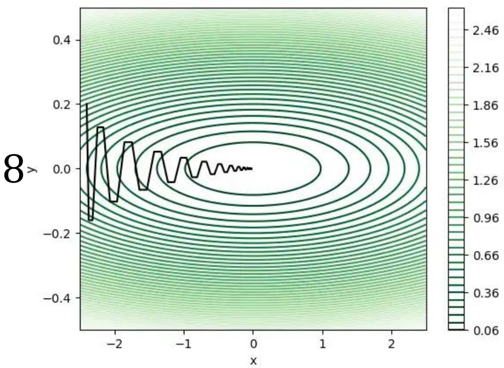
$x^{(k+1)} = x^{(k)} + z^{(k+1)}$

With $\beta = 0$, we obtain the gradient descent method.

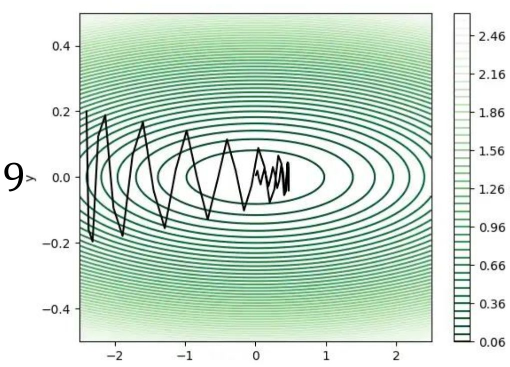
$\beta = 0$



$\beta = 0.8$



$\beta = 0.9$



Example Python code

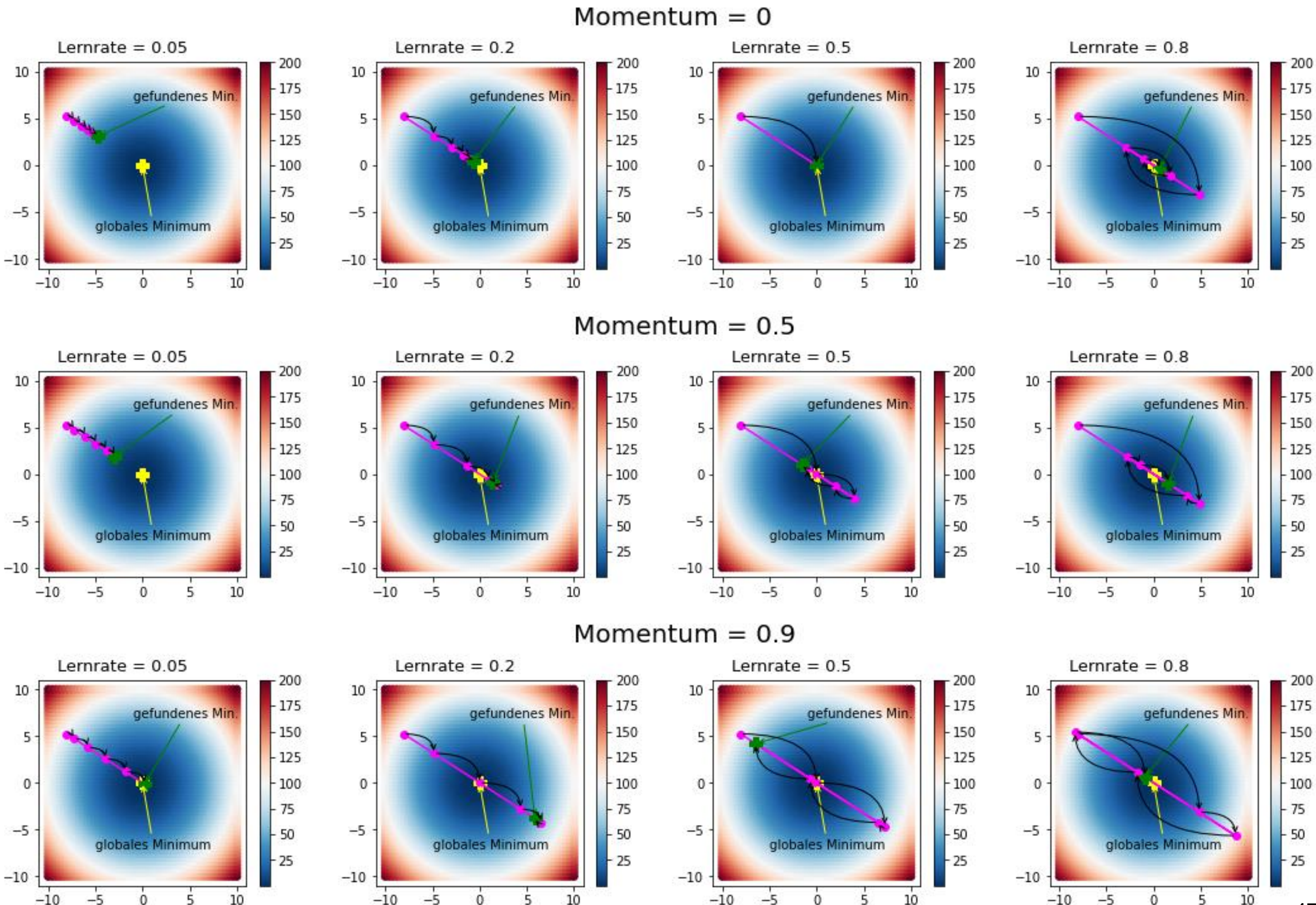
```
def gradient_descent(max_iterations,threshold,x_init,obj_func,grad_func,learning_rate=0.05, momentum=0.8):
    x = x_init
    x_history = x
    f_history = obj_func(x,extra_param)
    delta_x = np.zeros(x.shape)
    i = 0
    diff = 1.0e10
    while  i<max_iterations and diff>threshold:
        delta_x = -learning_rate*grad_func(x,extra_param) + momentum*delta_x
        x = x+delta_x
        # specify history of x and f
        x_history = np.vstack((x_history,x))
        f_history = np.vstack((f_history,obj_func(x,extra_param)))
        # update change in objective
        i+=1
        diff = np.absolute(f_history[-1]-f_history[-2])
     return w_history,f_history

def f(x):
    return np.sum(x*x)

def grad(x):
    return 2*x
...

gradient_descent(5,-1,x_init, f,grad,eta,alpha)
```

$$z^{(k+1)} = -\eta \nabla f\left(x^{(k)}\right) + \beta z^{(k)}$$
$$x^{(k+1)} = x^{(k)} + z^{(k+1)}$$


Momentum = 0
Lernrate = 0.05
Lernrate = 0.2
Lernrate = 0.5
Lernrate = 0.8
Momentum = 0.5
Lernrate = 0.05
Lernrate = 0.2
Lernrate = 0.5
Lernrate = 0.8
Momentum = 0.9
Lernrate = 0.05
Lernrate = 0.2
Lernrate = 0.5
Lernrate = 0.8
gefundenes Min.
globales Minimum

Detour: Stochastic Gradient Descent (SGD)

- Vanilla gradient descent in data analysis and machine learning is slow as the gradient is taken over all samples of the training data set.
- The Robbins-Monro (1950‘s) algorithm introduced stochastic gradient descent.
- Stochastic gradient descent takes a single sample or a small batch of samples to compute a (noisy) gradient estimate. The last term is a noise term.

$$x^{(k+1)} = x^{(k)} - \alpha \nabla f\left(x^{(k)}\right) - \eta \varepsilon^{(k)}$$
$$y^{(k)} = x^{(k)} - \alpha \nabla f\left(x^{(k)}\right)$$
$$y^{(k+1)} = x^{(k+1)} - \alpha \nabla f\left(x^{(k+1)}\right)$$
$$y^{(k+1)} = y^{(k)} - \eta \varepsilon^{(k)} - \alpha \nabla f(y^{(k)} - \eta \varepsilon^{(k)})$$
$$\mathbb{E}\left[y^{(k+1)}\right] = y^{(k)} - \alpha \nabla \mathbb{E}[f\left(y^{(k)} - \eta \varepsilon^{(k)}\right)]$$

- Each step is considerably faster.
- It is easier to escape saddle points or narrow local minima.

Use the gradient descent method to find the minimum of the following function: $f(x) = (x-2)^2$.

Start point: $x^{(0)} = 10$, step size: ½.